Laplace Transforms

CLO-II

Chapter#06 (Advance Eng. Mathematics)

Chapter#08 (section 8.1 & 8.2, Differential Equations)

Section 8.1

Definition of the Laplace Transform

The Laplace transform is a kind of *integral transform*. It takes a function $f(\mathbf{t})$, of one variable $\mathbf{t}$ (time) into a function $\mathbf{F(s)}$ of another variable s (frequency). Let $f(\mathbf{t})$ be a given function defined for all $\mathbf{t} \geq 0$. The Laplace transform of $f(\mathbf{t})$ is denoted by $\mathcal{L}\{f(t)\}$ or $\mathcal{L}\{f\}$, $\mathcal{L}$ being Laplace transform operator. It is defined as:

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t)\,dt = F(s). \qquad (1)$$

Linearity of the Laplace Transform

The Laplace transform is a linear operator. This property of the former is stated as under:

If f(t) and g(t) are any two functions having Laplace transforms, and if a and b are any constants, then,

$$\mathcal{L}\{a f(t) + b\, g(t)\} = a\,\mathcal{L}\{f(t)\} + b\,\mathcal{L}\{g(t)\}. \qquad (2)$$

Accordingly, we say that Laplace transform operator $\mathcal{L}$ is a linear operator.

Compute Laplace transform of following functions.

Sec # 8.1

Example # (01) $\boxed{f(t) = 1}$ — (1)

Sol

As,

Step (i) $\Rightarrow$ $L[f(t)] = \int_0^\infty e^{-st} f(t)\,dt$ — (2)

So, put (1) in (2)

Step (ii) $\Rightarrow$ (2) $\Rightarrow$ $L[1] = \int_0^\infty e^{-st} \cdot 1\, dt$ — (3)

Step (iii) $\Rightarrow$ $L[1] = \int_0^\infty e^{-st}\,dt = \left.\frac{e^{-st}}{-s}\right|_{t=0}^{t=\infty}$ — (4)

$= -\frac{1}{s}\left(e^{-\infty} - e^{-0}\right)$ — (5)

$= -\frac{1}{s}(0-1)$

Note:

$\boxed{\begin{array}{l} e^{-\infty} = 0 \\ e^{0} = 1 \end{array}}$

$\boxed{L[1] = \frac{1}{s}}$ — (6)

Example # (2)

$$\boxed{f(t) = t} \quad \text{--- (1)}$$

Sol

01 As

$$L[f(t)] = \int_0^\infty e^{-st} f(t)\, dt \quad \text{--- (2)}$$

02

$$L[t] = \int_0^\infty e^{-st}\, t\, dt \quad \text{--- (3)}$$

Apply integration by parts.

03

$$L[t] = t\,\frac{e^{-st}}{-s}\Big]_{t=0}^{t=\infty} - \int_0^\infty \left(\frac{e^{-st}}{-s}\right)(1)\, dt \quad \text{--- (4)}$$

$$L[t] = -\frac{1}{s}(0-0) + \frac{1}{s}\int_0^\infty e^{-st}\, dt \quad \text{--- (5)}$$

$$L[t] = \frac{1}{s}\left[\frac{e^{-st}}{-s}\right]_{t=0}^{t=\infty} \quad \text{--- (6)}$$

$$L[t] = -\frac{1}{s^2}\left(e^{-\infty} - e^{-0}\right) \quad \text{--- (7)}$$

$$\boxed{L[t] = \frac{1}{s^2}} \quad \text{--- (8)}$$

(2)

Example #03

$\boxed{f(t) = \sin wt}$ — (1)

w is a real number.

$\therefore$ $L[f(t)] = \int_0^{\infty} e^{-st} f(t)\, dt$ — (2)

$\therefore$ $L[\sin wt] = \int_0^{\infty} e^{-st} \sin wt \, dt$ — (3)

Method 01 : Apply integration by parts.

Method 02 is,

$$L[\sin wt] = Im\left[\int_0^{\infty} e^{-st} \cdot e^{iwt}\, dt\right] \quad (4)$$

bcoz, $\boxed{e^{iwt} = \cos wt + i \sin wt}$ — (5)

$Re[e^{iwt}] = \cos wt$ — (6), $Im[e^{iwt}] = \sin wt$ — (7)

$$\therefore L[\sin wt] = Im\left[\int_0^{\infty} e^{-st + iwt}\, dt\right] \quad (8)$$

$$= Im\left[\int_0^{\infty} e^{-t(s-iw)}\, dt\right] \quad (9)$$

$$= Im\left[\left[\frac{e^{-t(s-iw)}}{-(s-iw)}\right]_{t=0}^{t=\infty}\right] \quad (10)$$

$$= Im\left[-\frac{1}{(s-iw)}\left(e^{-\infty}-e^{-0}\right)\right] \quad (11)$$

$$= Im\left[\frac{1}{s-iw}\right] \quad (12)$$

$$= Im\left[\frac{1}{s-iw} \times \frac{s+iw}{s+iw}\right] \quad (13)$$

$$= Im\left[\frac{s+iw}{s^2+w^2}\right] \quad (14)$$

$$= Im\left[\frac{s}{s^2+w^2} + i\frac{w}{s^2+w^2}\right] \quad (15)$$

$$\boxed{L[\sin wt] = \frac{w}{s^2+w^2}} \quad (16)$$

Note:
if we consider real part of eq ( )
then $\boxed{L[\cos wt] = \frac{s}{s^2+w^2}}$ (17)

Example 04

$$\boxed{L[\cos\omega t] = \frac{s}{s^2+\omega^2}} \quad (*)$$

Done previously.

Example 05

$$\boxed{f(t) = e^{at}} \quad (1)$$

Sol

01 $$L[f(t)] = \int_0^\infty e^{-st} f(t)\,dt \quad (2)$$

02 $$L[e^{at}] = \int_0^\infty e^{-st} e^{at}\,dt \quad (3)$$

03 $$L[e^{at}] = \int_0^\infty e^{-t(s-a)}\,dt \quad (4)$$

$$= \left[\frac{e^{-t(s-a)}}{-(s-a)}\right]_{t=0}^{t=\infty} \quad (5)$$

$$= -\frac{1}{s-a}\left[e^{-\infty} - e^{-0}\right] \quad (6)$$

$$\boxed{L[e^{at}] = \frac{1}{s-a}} \quad (7)$$

Similarly:

$$\boxed{L[e^{-at}] = \frac{1}{s+a}} \quad \text{— (1)}$$

Exercise

$$\boxed{L[t^2] = \frac{2!}{s^3}} \quad \text{— (2)} \qquad \boxed{L[t] = \frac{1}{s^2}} \quad \text{— (3)}$$

$$\boxed{L[t^3] = \frac{3!}{s^4}} \quad \text{— (4)}$$

$$\vdots$$

$$\boxed{L[t^n] = \frac{n!}{s^{n+1}}} \quad \text{— (5)}$$

Examples (How to apply linearity property of Laplace Transform)

Example

Evaluate $\mathcal{L}\{\ \mathbf{cosh\ at}\ \}$ (1)

Solution

We know that **cosh at** $= \dfrac{e^{at}+e^{-at}}{2}$. (2)

$$\mathcal{L}\{\ \mathbf{cosh\ at}\ \} = \mathcal{L}\{\frac{e^{at}+e^{-at}}{2}\} \quad (3)$$

$$= \frac{1}{2}\,[\mathcal{L}\{e^{at}\} + \mathcal{L}\{e^{-at}\}] \quad (4)$$

$$= \frac{1}{2}\,[\frac{1}{s-a} + \frac{1}{s+a}] \quad (5)$$

$$= \frac{1}{2}\,[\frac{2s}{s^2-a^2}] \quad (6)$$

$$\mathcal{L}\{\ \mathbf{cosh\ at}\ \} = \frac{s}{s^2-a^2}\,. \quad (7)$$

Example

Evaluate $\mathcal{L}\{4 \sinh 3t - 18e^{-5t}\}$ (1)

Solution

$$\mathcal{L}\{4 \sinh 3t - 18e^{-5t}\} = 4\,\mathcal{L}\left\{\frac{e^{3t} - e^{-3t}}{2}\right\} - 18\,\mathcal{L}\{e^{-5t}\}. \quad (2)$$

Or,

$$\mathcal{L}\{4 \sinh 3t - 18e^{-5t}\} = 2\,[\,\mathcal{L}\{e^{3t}\} - \mathcal{L}\{e^{-3t}\} - 18\,\mathcal{L}\{e^{-5t}\} \quad (3)$$

$$= 2\left[\frac{1}{s-3} - \frac{1}{s+3}\right] - 18\left[\frac{1}{s+5}\right] \quad (4)$$

$$= 2\left[\frac{6}{s^2-9}\right] - \frac{18}{s+5} \quad (5)$$

Some Functions $f(t)$ and Their Laplace Transforms $\mathcal{L}(f)$

	$f(t)$	$\mathcal{L}(f)$		$f(t)$	$\mathcal{L}(f)$
1	1	$1/s$	7	$\cos \omega t$	$\dfrac{s}{s^2+\omega^2}$
2	t	$1/s^2$	8	$\sin \omega t$	$\dfrac{\omega}{s^2+\omega^2}$
3	t^2	$2!/s^3$	9	$\cosh at$	$\dfrac{s}{s^2-a^2}$
4	t^n $(n = 0, 1, \cdots)$	$\dfrac{n!}{s^{n+1}}$	10	$\sinh at$	$\dfrac{a}{s^2-a^2}$
5	t^a (a positive)	$\dfrac{\Gamma(a+1)}{s^{a+1}}$	11	$e^{at}\cos \omega t$	$\dfrac{s-a}{(s-a)^2+\omega^2}$
6	e^{at}	$\dfrac{1}{s-a}$	12	$e^{at}\sin \omega t$	$\dfrac{\omega}{(s-a)^2+\omega^2}$

EXERCISE – 8.1

Note: Answers are given against the respective questions within square brackets.

Use the definition of the Laplace transform to obtain the transform of $f(t)$, when $f(t)$ is given by:

1. $3t + 4$; $\left[\frac{3}{s^2} + \frac{4}{s}\right]$

2. $t^2 + at + b$; $\left[\frac{2}{s^3} + \frac{a}{s^2} + \frac{b}{s}\right]$

3. $(a + bt)^2$; $\left[\frac{a^2}{s} + \frac{2ab}{s^2} + \frac{2b^2}{s^3}\right]$

4. $\cos(\omega t + \theta)$; $\left[\frac{s\cos\theta - \omega\sin\theta}{s^2 + \omega^2}\right]$

5. $\sin(bt + a)$; $\left[\frac{b\cos a + s\sin a}{s^2 + b^2}\right]$

6. $\sin^2 t$; $\left[\frac{1}{2s} - \frac{s}{2s^2 + 8}\right]$

7. $\cos^2 t$; $\left[\frac{1}{2s} + \frac{s}{2s^2 + 8}\right]$

8. $\sin^2 \omega t$; $\left[\frac{1}{2s} - \frac{s}{2s^2 + 8\omega^2}\right]$

9. $\cos^2 \omega t$; $\left[\frac{1}{2s} + \frac{s}{2s^2 + 8\omega^2}\right]$

10. e^{3t+4}; $\left[\frac{e^4}{s-3}\right]$

<u>**Theorem (The First Shift Theorem)**</u>

If $f(t)$ is a function having Laplace transform $\mathbf{F(s), s > 0}$, then,

$$\mathcal{L}\,\{e^{at} f(t)\} = F\,(s - a),\;\; s > a \quad (1)$$

Ex#01 Find Laplace transform of

$$\boxed{f(t) = te^{-2t}} \quad \text{— (1)}$$

Solution $\boxed{\text{Method \#01}}$

By definition of Laplace transform,

$\boxed{\text{Step \#01}} \Rightarrow$

$$\boxed{\mathcal{L}\{t e^{-2t}\} = \int_0^{\infty} e^{-st}(te^{-2t})\,dt} \quad \text{— (2)}$$

$$= \int_0^{\infty} t e^{-(s+2)t}\,dt \quad \text{— (3)}$$

$\boxed{\text{Step \#03}} \Rightarrow$

$$= \left.\frac{t e^{-(s+2)t}}{-(s+2)}\right|_0^{\infty} - \int_0^{\infty} \frac{e^{-(s+2)t}}{-(s+2)}\,dt \quad \text{— (4)}$$

$\boxed{\text{Step \# 04}} \Rightarrow$

$$= 0 + \frac{1}{s+2}\left[\frac{e^{-(s+2)t}}{-(s+2)}\right]_0^{\infty} \quad \text{— (5)}$$

$$= \frac{1}{(s+2)^2}, \quad \text{— (6)}$$

Method #02 (First shift theorem)

$$L[t e^{-2t}] = ?? \quad \text{— (1}$$

01 As we know, if $\boxed{L[f(t)] = F(s)}$, — (2 and

$$\boxed{L[e^{at} f(t)] = F(s-a)} \quad \text{— (3}$$

02 To apply first shift theorem, compare L.H.S of both (1 and (3,

$$\boxed{f(t) = t} \quad \text{— (4}$$

$$\Rightarrow L[f(t)] = L[t] \quad \text{— (5}$$

$$\Rightarrow \boxed{F(s) = \frac{1}{s^2}} \quad \text{— (6}$$

03 (Use First shift / replace s by s−a (s → s−a))

(5 $\Rightarrow L[e^{-2t} f(t)] = L[e^{-2t} t]$

$$= \frac{1}{(s+2)^2}$$

(replace just $s \rightarrow s+2$ ($\because e^{-2t}$) in (6,)

04

$$\boxed{L[t e^{-2t}] = \frac{1}{(s+2)^2}}$$

Ex

$L[t^2 e^{2t}] = ??$ — (1

Sol

01

$$\boxed{L[f(t)] = F(s)} \quad \text{— (2}$$

$$\boxed{L[e^{at} f(t)] = F(s-a)} \quad \text{— (3}$$

(Just replace $s \to s-a$ in (2)

02 (Determine $f(t)$)

Compare L.H.S of (1 and (3

$$\Rightarrow \boxed{f(t) = t^2} \quad \text{— (4}$$

$$L[f(t)] = L[t^2] \quad \text{— (5}$$

$$\boxed{F(s) = \frac{2}{s^3}} \quad \text{— (6}$$

03 (use first shift / replace s by $s-a$)

$$\boxed{L[e^{2t} t^2] = \frac{2}{(s-2)^3}} \quad \text{— (7}$$

(Just replace $s \to s-2$ in (6)

EXERCISE 8. 2

Note: Answers are given against the respective questions within square brackets.

Obtain the Laplace transform of the following functions.

1. $t\,e^{kt}$; $\left[\dfrac{1}{(s-k)^2}\right]$

2. $e^{t}\sin t$; $\left[\dfrac{1}{(s-1)^2+1}, \text{i.e., } \dfrac{1}{s^2-2s+2}\right]$

3. $e^{-3t}\sin 2t$; $\left[\dfrac{2}{(s+3)^2+4}, \text{i.e., } \dfrac{2}{s^2+6s+13}\right]$

4. $e^{-kt}\sin at$; $\left[\dfrac{a}{(s+k)^2+a^2}\right]$

5. $e^{-kt}\cos at$; $\left[\dfrac{s+k}{(s+k)^2+a^2}\right]$

6. $e^{-t}\sin(\omega t+\theta)$; $\left[\dfrac{\omega\cos\theta+(s+1)\sin\theta}{(s+1)^2+\omega^2}\right]$